

CALVIN NGUYEN

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Education

North Carolina State University
Computer Science, PhD: Anticipated December 2027
Computer Science, MCS: May 2024
Computer Science, BS: May 2019

Work Experience

- Natrx - Data Scientist, Geospatial ML & Remote Sensing: (Aug 23 - Jul 26)
- Developed a geospatial machine-learning pipeline for land/water classification and shoreline-change detection, supporting analysis of 2,909 shoreline miles across 39 North Carolina study areas and 93,017 measurement locations.
 - Developed and validated Guyana's PlanetScope-based shoreline classification workflow, achieving 97.9–99.2% accuracy across 300 bootstrapped evaluations and enabling erosion and accretion mapping along 682 km of coastline.
 - Created an experimental submerged-mudbank detection method using Canny edge detection and support vector machines on 50 cm satellite imagery to identify wave-damping and wave-breaking signatures associated with fluid and consolidated mud.
 - Produced land/water segmentation for Biscayne Bay erosion analysis and assembled a property-level geospatial dataset enriched with ownership, zoning, land use, address, and permitting-related attributes.
 - Evaluated eight physical datasets for Billion Oyster Project across six quality and integration criteria; implemented literature-derived rules for depth, velocity, wave exposure, erosion, and infrastructure proximity within a prioritization analysis of 78 candidate restoration sites.
 - Processed topo-bathymetric LiDAR, water-level datums, SAV, and soil-carbon data for coastal restoration planning; derived shoreline elevations, slopes, and cross-sections and produced graphics supporting preliminary design and permitting.
- Wake County Library - Page: (June 23 - Aug 23)
- Sort and shelve library materials in alphabetic and numerical order.
 - Check in library materials using the library's automated computer system.
 - Box and unbox library materials and assist library members with book donations.

Research

Foundation Models for Semantic Segmentation of Thick/Thin Clouds and Cloud Shadows: A Comparative Study

| ACM SIGSPATIAL 2025 Short Paper

- Benchmarked HQ-SAM, Prithvi, SatlasNet, LSKNet/UNetFormer, and FMask on L8-Biome for thick cloud, thin cloud, and cloud shadow segmentation.
- Reworked model implementations for TorchScript compatibility and built reproducible PyTorch pipelines for data preparation, training, and performance evaluation.

Geospatial Foundation Models for Cloud Segmentation | Submitted to IEEE DSAA 2026

- Built a large-scale benchmark of 16 geospatial foundation models across 210,519 Landsat and Sentinel-2 images from four cloud-masking datasets.
- Engineered a reproducible PyTorch Lightning/Hydra training suite with serialized splits, controlled randomness, and configurable segmentation heads.

Advisor: Ranga Raju Vatsavai, Professor, North Carolina State University

Volunteering

- JC Raulston Arboretum: (Spring/Summer 2023)
- Audit plant locations and labels from bed maps
 - Replaced stakes and labels for plants
 - Consulted on updating deprecated database connector
 - Demonstrated usage of RESTful APIs to connect to Postgres/GIS

Technical Skills

Python, pandas; GIS, Remote Sensing, Spatial Analysis; GeoPandas, Rasterio/GDAL; scikit-learn, XGBoost, Computer Vision